

Replication Package

Fillon & Guivarch — *The Need for Regulation of Climate Subsystems*

Abstract

This document describes the replication package for Fillon & Guivarch, *The Need for Regulation of Climate Subsystems*. It details the contents of every folder, the role of each script and data file, and the order in which the pipeline must be executed to reproduce all results, tables, and figures. The package is organised in two top-level directories: `analysis_figures/` (econometric calibration and figure production, in R) and `numerical_model/` (numerical solution of the optimal intertemporal program, in Python).

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1 Overview of the package

The replication package is structured as follows:

`analysis_figures/`

R scripts for the econometric calibration of Amazon dynamics, the Bayesian calibration against expert elicitations, and the production of all figures and tables in the paper. Contains three subfolders: `data/`, `output/`, and `figure/`.

`numerical_model/`

Python codebase for solving the stochastic dynamic optimisation problem and simulating the calibrated model. Contains four subfolders: `model/`, `parameters/`, `run/`, and `outputs/`.

A high-level description of the pipeline is given in Section 5; the contents of each subfolder are detailed in Sections 3–4.

2 Data availability

Most data files are stored in `analysis_figures/data/`. Three datasets are too heavy to be distributed in the package and must be downloaded separately:

- `rainfall/` — monthly precipitation rasters from ISIMIP 2B, used to compute the Maximum Cumulative Water Deficit (MCWD) anomaly.
- `temperature/` — daily temperature rasters from ISIMIP 2B, used to compute annual mean temperature.
- `earthdata_carbon2010/` — above- and belowground biomass carbon stock rasters at ~300 m resolution, used as regression weights.

These datasets are available either from the project [Google Drive](#) or directly from [ISIMIP](#) and [EarthData](#). They must be placed in `analysis_figures/data/` under the names listed above.

3 Folder `analysis_figures/`

This folder contains the R pipeline producing the calibration of Amazon dynamics and all figures/tables of the paper. It is organised into three subfolders described below.

3.1 `analysis_figures/data/`

Contains all input data used by the R scripts. Beyond the three heavy datasets listed above, this folder includes:

`subregions_souza2016/`

Shapefile delineating the boundaries of the Amazon biome and its subregions, from Souza et al. (2016). Used to crop and mask climate rasters, to define regional fixed effects, and to compute the total area of the rainforest.

`temperature_ERF/`

Effective Radiative Forcing (ERF) trajectories for the SSP scenarios (SSP1-1.9, SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP4-3.4, SSP4-6.0, SSP5-3.4-OS, SSP5-8.5), as provided by IPCC AR6 WG1 (Chapter 7 supplementary material). Used to derive global mean temperature pathways via a two-box energy balance model.

`kriegler_expertproba.csv`

Expert elicitation data from Kriegler et al. (2009): for each expert and for each warming corridor (*low*, *medium*, *high*), lower and upper bounds on the probability of partial Amazon dieback. Used in the Bayesian calibration to build the target distributions.

`temp_ssp_ipcc2200.csv`

Five-year-step temperature anomaly trajectories for the SSP scenarios, derived from the ERF data, scaled to reproduce the central path of Kriegler corridors and converted to regional Amazon warming using a regional-to-global TCRE ratio. Used both as input to the calibration and as forcing for SSP-based simulations.

3.2 `analysis_figures/output/`

Contains intermediate outputs and result files produced by the R scripts.

temperature/

One GeoTIFF per (climate model, RCP) combination storing annual mean temperature over the Amazon, computed from October of year n to September of year $n + 1$, in °C. Produced by `estimation.R`.

rainfall/

One GeoTIFF per (climate model, RCP) combination storing the annual MCWD anomaly relative to the 1981–2004 baseline, in mm. Produced by `estimation.R`.

final_dataframe.csv

Cell-level panel dataset combining temperature, MCWD anomaly, carbon weight, and Amazon subregion identifier, for all (model, RCP, year, cell) combinations. Used as the regression input.

summary_stat.tex

LaTeX table of descriptive statistics (sample size, mean and standard deviation of temperature and MCWD anomaly) by climate model and RCP.

test_fe_table.tex, panel_spec_tests.tex, error_quality_tests.tex

LaTeX tables reporting panel diagnostic tests (F test for fixed effects vs pooled OLS, Hausman test, F test for time fixed effects, Breusch–Pagan, Wooldridge serial correlation, Pesaran cross-sectional dependence).

results_full.tex

LaTeX table reporting the regression coefficient of carbon losses on log-transformed temperature anomaly for all specifications (FE, FE + time FE, FE + regional FE, FE + time + regional FE, weighted) and all climate models, with classical and Driscoll–Kraay standard errors.

table1_main_results.tex

LaTeX table reporting the preferred specification across all RCPs combined, by climate model.

table2_rcp_results.tex

LaTeX table reporting the preferred specification by RCP and climate model.

fe_table.tex

LaTeX table grouping all regression results by model, RCP and specification.

epsilons.csv

Mapping coefficient ε converting log temperature anomaly into share of carbon losses, for each climate model and for the multi-model average. Read by `calibration.R`.

good_candidates_step1.csv

Subset of LHS candidates that fall within the credible intervals of all three Kriegler corridors, ordered by best (smallest) credible-interval shape.

candidates_log.csv

Detailed log of every LHS candidate that achieved 3/3 corridors in bounds, with simulated probabilities, deviations from expert means and MAE.

sensitivity_sets_final.csv

Three calibrated parameter sets (*baseline*, *robust-low*, *robust-distant*) used for sensitivity analysis in the welfare module.

calibration_kriegler.tex

LaTeX table summarising the calibrated $(\alpha, \beta, g_0, \beta_0, \eta, \Upsilon)$ parameters for each of the three calibration sets.

3.3 analysis_figures/figure/

Contains the figures produced for the paper (PDF and PNG). The contents are not detailed individually, as each figure corresponds to a specific figure of the manuscript.

3.4 R scripts

calibration_dynamics/estimation.R

Step 1A. Loads ISIMIP 2B temperature and precipitation rasters for four climate models (GFDL, HadGEM, IPSL, MIROC) and three RCPs (2.6, 6.0, 8.5), computes annual mean temperature and annual MCWD anomaly, builds a cell-level panel dataset, runs panel diagnostic tests and fixed-effect regressions of carbon losses on log temperature anomaly, derives the climate-shock coefficient ε for each model, and produces all associated descriptive tables. Also derives SSP-consistent temperature trajectories from the IPCC AR6 ERF data through a two-box energy balance ODE.

calibration_dynamics/calibration.R

Step 1B. Reads the climate-shock coefficient produced by Step 1A and the expert elicitation data of Kriegler et al. (2009). Fits a Beta distribution to each warming corridor (low, medium, high) by method of moments. Calibrates the parameters of the Amazon dynamics ($\mu_\varepsilon, \phi_\varepsilon, g_0, \eta, \beta_0, \Upsilon$) via a Latin-Hypercube Sampling exploration combined with bisection on the width of the credible interval. Selects three parameter sets used in the welfare analysis (benchmark and two counterfactuals) and exports them to **numerical_model/parameters/**.

plot_results.R

Step 3. Reads outputs from **numerical_model/outputs/** for the runs of the benchmark and the two counterfactuals, and produces all final figures of the paper, including the decomposition of the SCC into its temperature, Amazon, feedback and interaction components.

4 Folder numerical_model/

This folder contains the Python implementation of the stochastic dynamic optimisation problem, its solution by Chebyshev value-function interpolation, and the simulation of stochastic paths.

4.1 numerical_model/model/

Contains the Python source code defining the economic model (production function, abatement cost, damage function, transition equations of the state variables) as well as the numerical-solution machinery (Chebyshev basis construction, value function iteration, policy function extraction). This folder is the implementation of the model described in the paper and is shared by all runs.

4.2 numerical_model/parameters/

Contains all parameter files read by the numerical model.

calibration_kriegler.csv

Calibrated parameters of the Amazon dynamics, produced by **calibration.R**. Contains three rows, one per calibration set (*baseline*, *robust-low*, *robust-distant*, identified by **calibration_id** $\in \{1, 2, 3\}$). Columns: **beta_alpha** and **beta_beta** (parameters of the Beta distribution governing the climate shock ε), **growth0** (g_0 , baseline growth rate of the Amazon), **beta0** (carrying capacity scaling parameter), **eta** (curvature of the feedback term), and **Upsilon** (intensity of the climate–Amazon feedback).

param_model.csv

Full set of parameters of the stochastic DICE-like model: rate of pure time preference, elasticity of intertemporal substitution, parameters of the production function, abatement cost function, damage function, climate sensitivity, transient climate response to cumulative emissions (TCRE), regional-to-global temperature scaling, initial conditions for capital, temperature, cumulative emissions, and the Amazon state, total area of the rainforest, deforestation-to-degradation ratio, and the temperature step ΔT used in the discretisation. Format: three columns (**variable**, **value**, **description**).

param_scenariosEU.csv

Average exogenous deforestation pathway used as the baseline rate of forest loss $\kappa(t)$ in the Amazon transition equation. One value per five-year period.

param_gen.csv

Numerical-solver settings: convergence tolerance for value function iteration, maximum number of iterations, Chebyshev polynomial degree along each state dimension, number of collocation nodes, time horizon and step, and seed for the random number generator used in the stochastic simulation.

4.3 numerical_model/run/

Contains the orchestration scripts and configuration files defining which runs to execute and with which specifications.

study.csv

Configuration file specifying the run number to be treated by the interpolation script. To launch a new run, the user edits this file with the desired run identifier; `meta_run1_interpolation.py` reads it and proceeds accordingly.

final_amazon_tcre.csv

Master list of all runs to be executed, with their full specifications. Each row corresponds to one run (`run0001`, `run0002`, ...) and encodes: the calibration set used (1, 2 or 3, mapping to baseline / counterfactual 1 / counterfactual 2), whether temperature damages are activated, whether the Amazon module is included, and the level of stochasticity (0 = deterministic, 1 = one risk on dT/dS , 2 = two risks on dT/dS and dA/dT). The benchmark calibration corresponds to runs 1–6, the first counterfactual to runs 7–11, and the second counterfactual to runs 12–16, following the run ordering described in Section 5.

meta_run1_interpolation.py

Reads `study.csv` to identify the target run, loads the corresponding specification from `final_amazon_tcre.csv` and the parameters from `numerical_model/parameters/`, builds the Chebyshev state space, performs value function iteration until convergence, and writes the Chebyshev coefficients of the value function (and other run-specific outputs) to `numerical_model/outputs/`.

meta_run2_simulation.py

For runs with stochastic shocks (runs 4–6, 9–11, 14–16), reads the Chebyshev coefficients produced by `meta_run1_interpolation.py` and simulates a large number of stochastic trajectories of the optimal program. Computes statistics (means, percentiles) and the various social-cost-of-carbon decompositions used in the paper.

4.4 numerical_model/outputs/

Contains one subfolder per run, named `final_amazon_tcre_run00XX/`. Each subfolder contains:

coef_V.csv

Chebyshev coefficients of the converged value function on the run-specific state space.

u_min.csv, u_max.csv

Lower and upper bounds of the state space used to build the Chebyshev basis (one row per state dimension). Note that runs that do not include the Amazon module reuse the state space defined when run 2 is executed; this is why run 2 must be executed before runs that share its state space.

param_model.csv, param_gen.csv

Snapshot of the parameter files used for this run, kept inside the run folder for full reproducibility.

control_V_notstochastic.csv

Optimal control trajectory (abatement rate) along the deterministic optimal path. Produced for deterministic runs (`stochastic = 0`, runs 1–3, 7–8, 12–13).

state_V_notstochastic.csv

Optimal state trajectory (capital, temperature, Amazon coverage) along the deterministic optimal path. Produced for deterministic runs.

outputs_stochastic.csv

Stochastic trajectories and welfare decomposition produced by `meta_run2_simulation.py` for stochastic runs (`stochastic` $\in \{1, 2\}$, runs 4–6, 9–11, 14–16). Contains, period by period, the optimal control, the three state variables, and the Social Cost of Deforestation (SCD) and Social Cost of Carbon Discounted Stochastic (SCCDS), each decomposed into temperature, Amazon subsystem, cross and covariance components.

outputs_stochastic5.csv, outputs_stochastic95.csv

Fifth and ninety-fifth percentiles across stochastic simulations of the control and state variables, used for confidence intervals in the figures.

5 Pipeline: order of execution

The complete pipeline runs in three steps. Each step depends on the previous one and must be executed in order.

Step 1 — Calibration of Amazon dynamics

Step 1A. Run `analysis_figures/calibration_dynamics/estimation.R`. This script ingests temperature and precipitation data from ISIMIP 2B, computes the climate-shock coefficient ϵ for each climate model, and writes `epsilons.csv` together with descriptive tables and panel diagnostic tables.

Step 1B. Run `analysis_figures/calibration_dynamics/calibration.R`. This script uses the coefficients produced in Step 1A to match the expert elicitations of Kriegler et al. (2009). Three calibration sets are produced: one benchmark and two counterfactuals. The output is written to `numerical_model/parameters/calibration_kriegler.csv`.

Step 2 — Solving the optimal intertemporal program

Six core specifications are run as the benchmark, then replicated under the two counterfactual calibrations:

Run 1

Deterministic (`stochastic = 0`), no temperature damages, exogenous paths of state variables without optimal control.

Run 2

Deterministic with temperature damages, without Amazon module.

Run 3

Deterministic with temperature damages, with Amazon module.

Run 4

Stochastic (`stochastic = 1`, single risk on dT/dS), with temperature damages, without Amazon.

Run 5

Stochastic (`stochastic = 1`), with damages, with Amazon.

Run 6

Stochastic (`stochastic = 2`, two risks on dT/dS and dA/dT), with damages, with Amazon.

The two counterfactual blocks repeat the same six specifications under alternative calibrations:

Runs 7–11

First counterfactual — alternative calibration from expert elicitations.

Runs 12–16

Second counterfactual — alternative calibration from expert elicitations.

Step 2A — Interpolation of the value function. Open `study.csv` and set the run number to be treated. Run 1 must be executed first, immediately followed by run 2: the state space used in all subsequent runs is built when run 2 is executed. Then run `meta_run1_interpolation.py`. Repeat for every run number, advancing through runs 3, 4, ..., 16 (with run 2 again as a prerequisite for counterfactual blocks if the state space is rebuilt).

Step 2B — Simulation of stochastic paths. For runs with stochasticity (runs 4–6 in the benchmark; runs 9–11 in the first counterfactual; runs 14–16 in the second counterfactual), open `meta_run2_simulation.py` once `meta_run1_interpolation.py` is complete for the corresponding run, set the run number, and execute `meta_run2_simulation.py`. This produces `outputs_stochastic.csv`, `outputs_stochastic5.csv` and `outputs_stochastic95.csv` inside the run folder.

Step 3 — Figures and tables of the paper

Once all 16 runs have completed Step 2A (and Step 2B for the stochastic ones), run `analysis_figures/plot_res`. This script reads the outputs of every run, computes the welfare statistics reported in the paper, and writes all final figures to `analysis_figures/figure/`. Selecting between the benchmark, counterfactual 1 or counterfactual 2 set of figures is controlled by the `id_scenario` variable at the top of the script.

End of replication package documentation.